METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET V-201

V-201 · HP separator

DOCUMENT NO.	:	MEOH-V1-ME-DS-002
DOCUMENT TITLE	:	Datasheet V-201
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

V-201 · HP separator

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-002.

General		
Service	Separates crude methanol from the loop gas; holds the loop gas inventory	MEOH-PID-002
Type	Vertical, mesh demister, liquid boot section	
Quantity	1	
Operating (model, design point)		
Pressure / temperature	75 bar g / 40 °C	PIC-203 SP 75, TI-253
Vapour out (recycle + purge)	1667 kNm³/h, M 5.83, ρ 16.5 kg/m³	model
Crude methanol out (FI-209)	131.5 t/h, ρ 814 kg/m³	LIC-204
Level NLL / alarms / trip	50 % / L 25 · H 75 % / LSHH-204 90 %	ALM, K.tripL
Design		
Design pressure	88 bar g	Zone A = PSV-201 set
Design temperature	120 °C	Loss of most E-203 cooling water (75 % less flow; design case) + 30 °C
MDMT	−29 °C	assumed; blowdown study TBC (EQ-em-05)
Code	ASME VIII Div. 1 (assumed)	
Corrosion allowance	3 mm	wet CO_2 / crude methanol
PWHT / impact test	Yes / yes	thickness
Hydrotest	116.4 bar g	UG-99(b)
Sizing		
Souders–Brown K	0.077 m/s	mesh pad with pressure correction (assumed)
Allowable vapour velocity	0.535 m/s	$u = K\sqrt{(\rho L - \rho G)/\rho G}$
Actual vapour flow	7.29 m³/s	
Required / selected ID	4165 / 4200 mm	
Level span (0–100 %)	2.17 m = 30 m³	K.Vboot
Tangent-to-tangent	5500 mm	LLLL 0.3 + span + 0.6 + inlet 0.9 + 1.0 + demister + 0.3 m (assumed)
Materials and thickness		
Shell, heads	Carbon steel SA-516 Gr 70 (N)	
Shell thickness	139.4 + 3 → 145 mm	UG-27(c)(1)
Head thickness	135 + 3 → 140 mm	UG-32(d)
Demister	SS 316 mesh, 150 mm	
Weight empty (pressure parts +10 %)	≈ 143 t	
Nozzles (ASME B16.5 Class 600, assumed)		
N1	DN900	Inlet from E-203 (line number not on the P&ID, EQ-ep-06)
N2	DN900	Vapour outlet · DN900-PG-205 (purge DN150-PU-203 branches off)
N3	DN200	Crude methanol outlet · DN200-CM-209 → LV-204
N4	DN150	PSV-201 inlet (set 88 bar g)
N5	DN100	HV-205 blowdown (restriction orifice, see RLB)
N6/N7	DN50	LT-204 bridle; LSHH-204
N8	DN50	PT-203 / PSHH-203
N9	DN50	TI-253
M1	DN600	Manway
Notes		
1	PSV-201 protects zones A and B of the loop; see MEOH-V1-PE-RLB-001.	

Notes		
2	Total cooling-water loss heats V-201 above its design temperature in the model (EQ-em-04).	